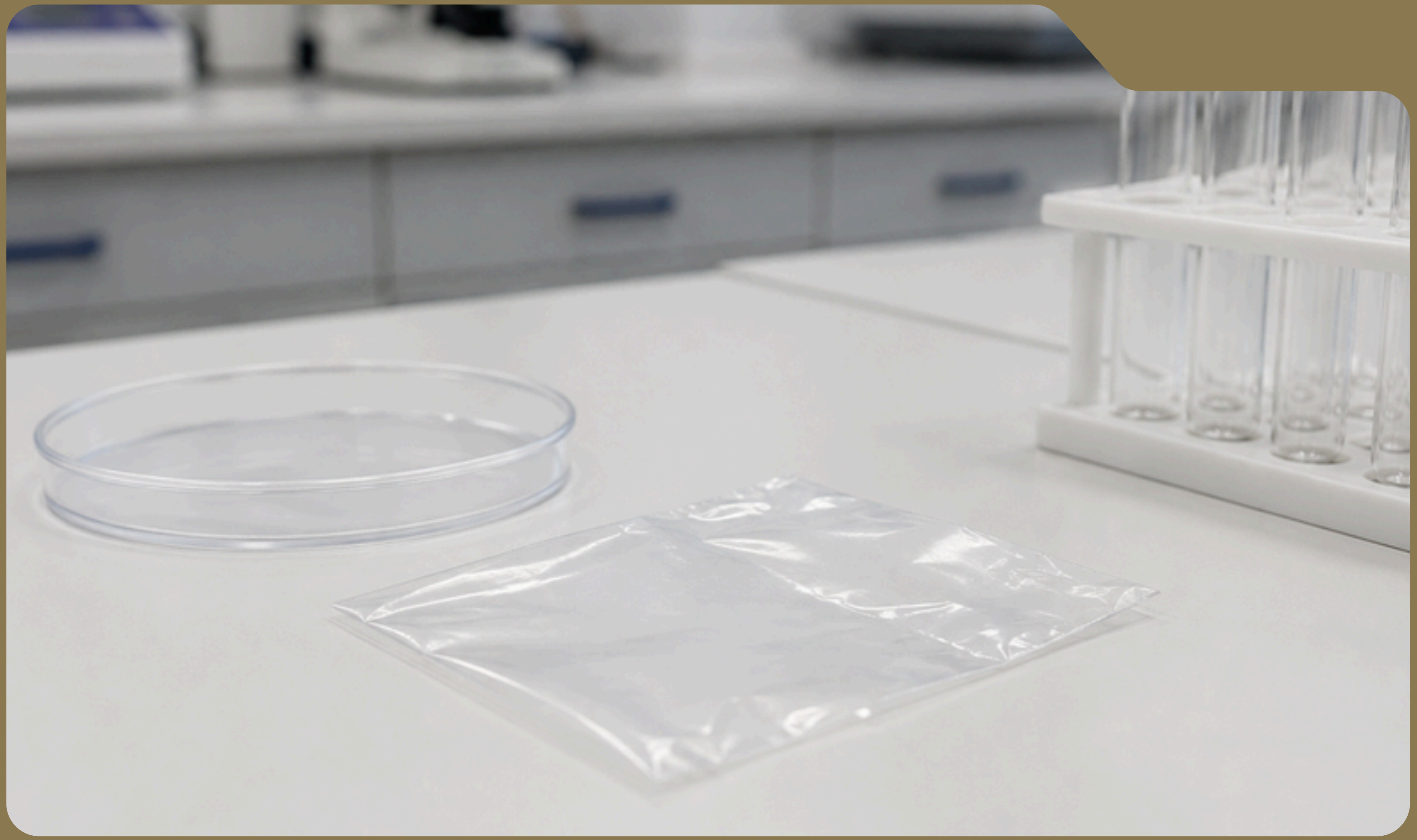
# SOLARSAFE Interim Report
# Materials Team

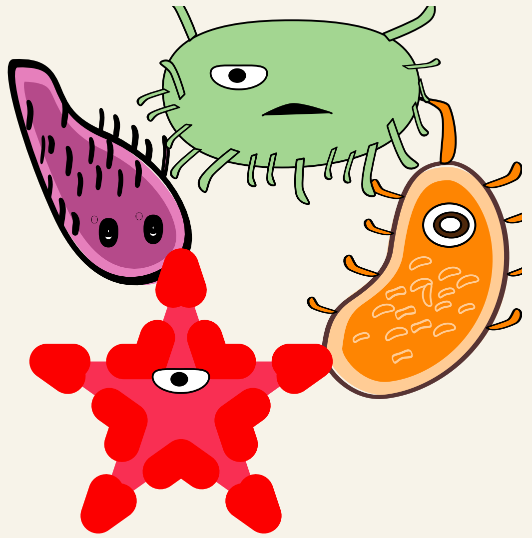


**Presented by**

Kunran Cui, Mirha Kashif


**CONTEXT - Aflatoxin Contamination**

Internal
• High moisture + oxygen → Aspergillus growth
• Respiration → heat generation → hot spots
• Hot spots → increased aflatoxin production
• Aflatoxins accumulate locally rather than uniformly (heterogeneous contamination)
• Lower moisture/$O_2$ and uniform conditions reduce risk



External
• Soil contact introduces spores and moisture
• Material degradation allows contaminant ingress
• High permeability increases moisture and oxygen exposure
• Barrier integrity reduces contamination risk

Goal
• Low moisture • Low oxygen • Uniform conditions • Fewer hot spots

Reference: Whitaker, T.B. et al. Aflatoxin Contamination Distribution Among Grains and Nuts. Journal of Food Protection. 1998.

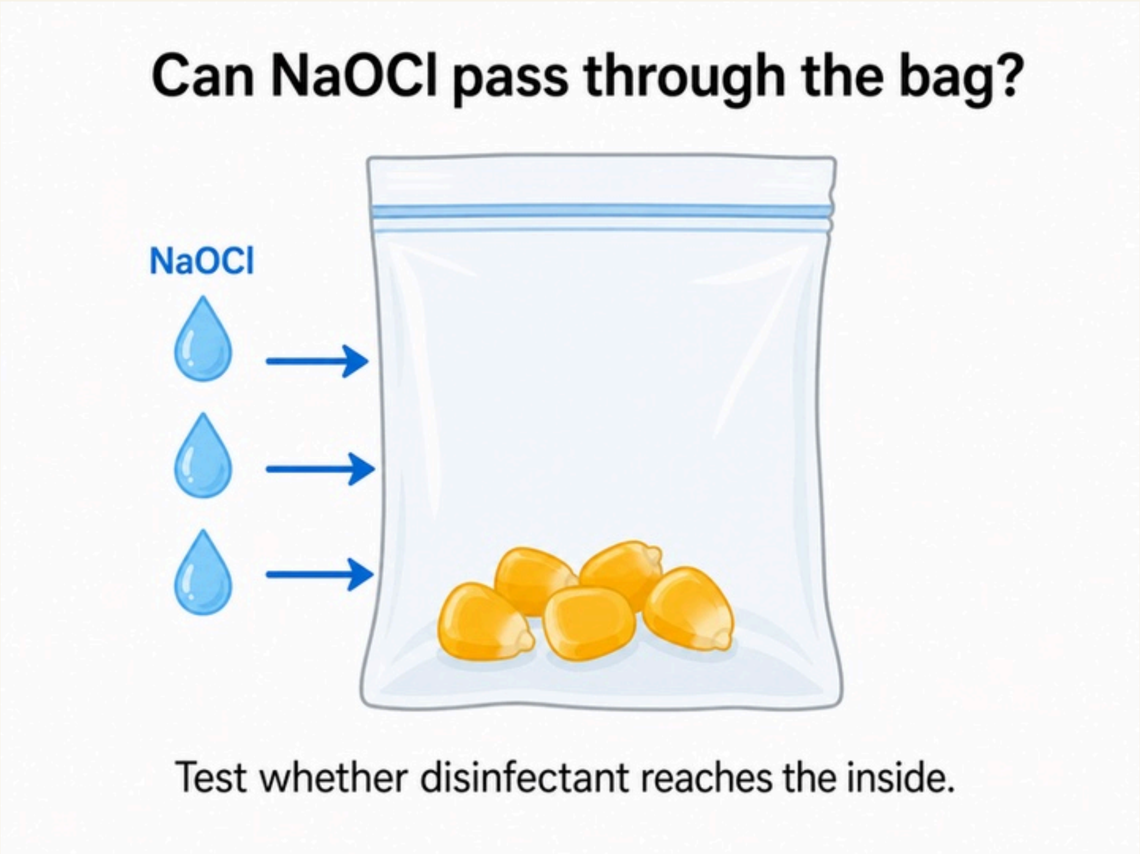
# Existing Problems

1. Sodium Hypochlorite-surface interaction - Experiments

Why use NaOCl?

Dilute NaOCl is commonly used as a disinfectant to reduce fungal contamination and can help **reduce aflatoxin-related risk**.

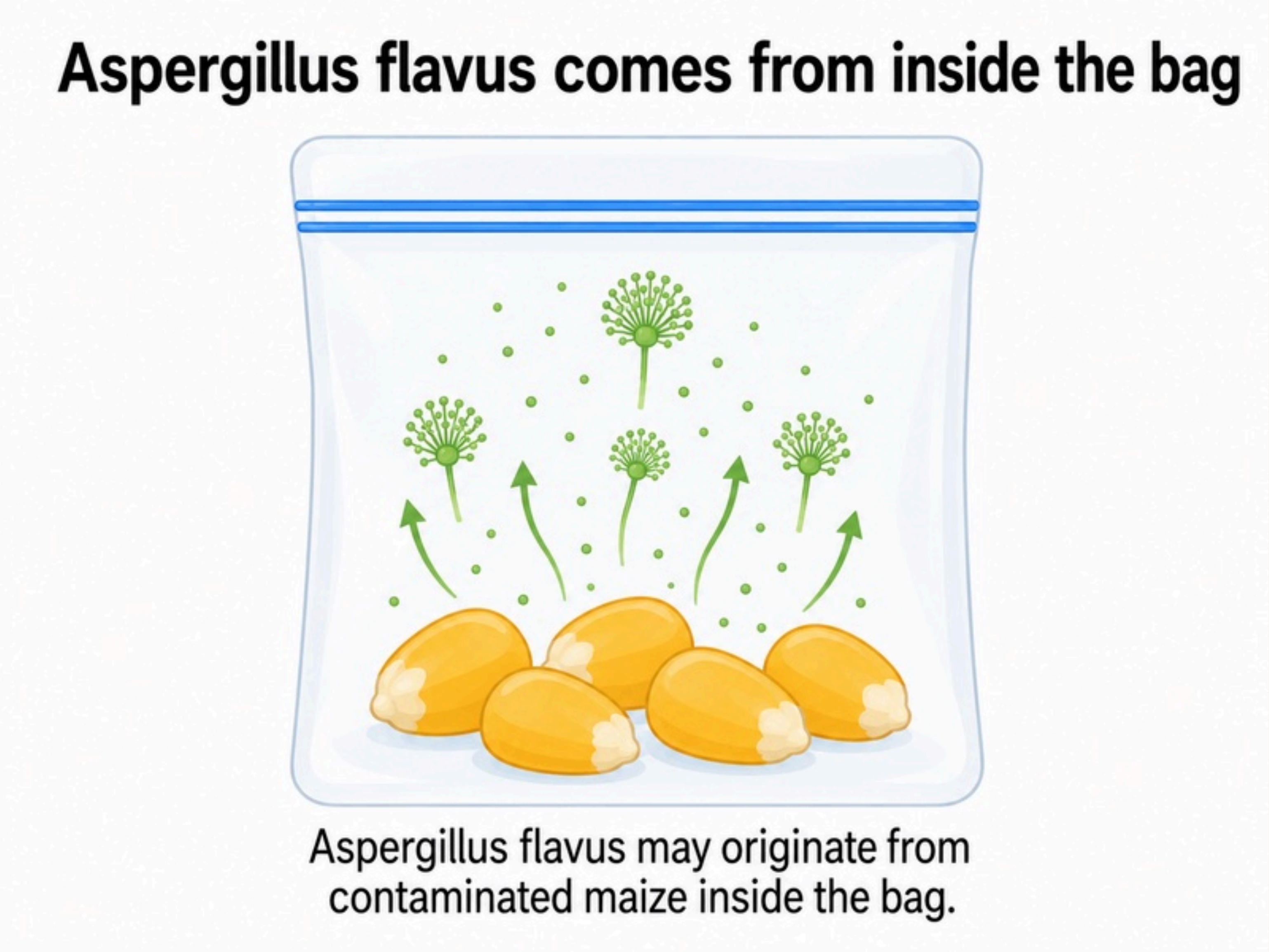
**Key Question: Does sodium hypochlorite degrade the drying plastic bag?**

# Existing Problems

2. Aspergillus flavus-surface interaction
   Research+Modelling

**Key Question: Does aspergillus flavus stick to the drying bag, so bags can't be reused?**



# Current Progress

### May 19th - 22nd

- **Contacting** labs/academics/literature reviews (emailed over 20 different people/labs/dep)

### May 23th - 25nd

- **Meetings** with academics & experimentalists: Rich, Gideon, Laura, etc.

### May 25th - 28th

- **Risk assessments, planning orders,** contact & meetings

### May 28th - 29th

- **Preparing for experiments** to be done in plastic lab: materials, set-up, procedure

### May 30th - 31th

- **Narrow down scope of materials**, see in later page

# Issues and Solutions

1. Sodium Hypochlorite-surface interaction - Experiments

   Problem: **Difficult to conduct chemical exposure test** without supervision

   - Reached out to materials, Professor Eugene Terentjev has agreed to supervise us

2. Aflatoxin-surface interaction - Research+Modelling

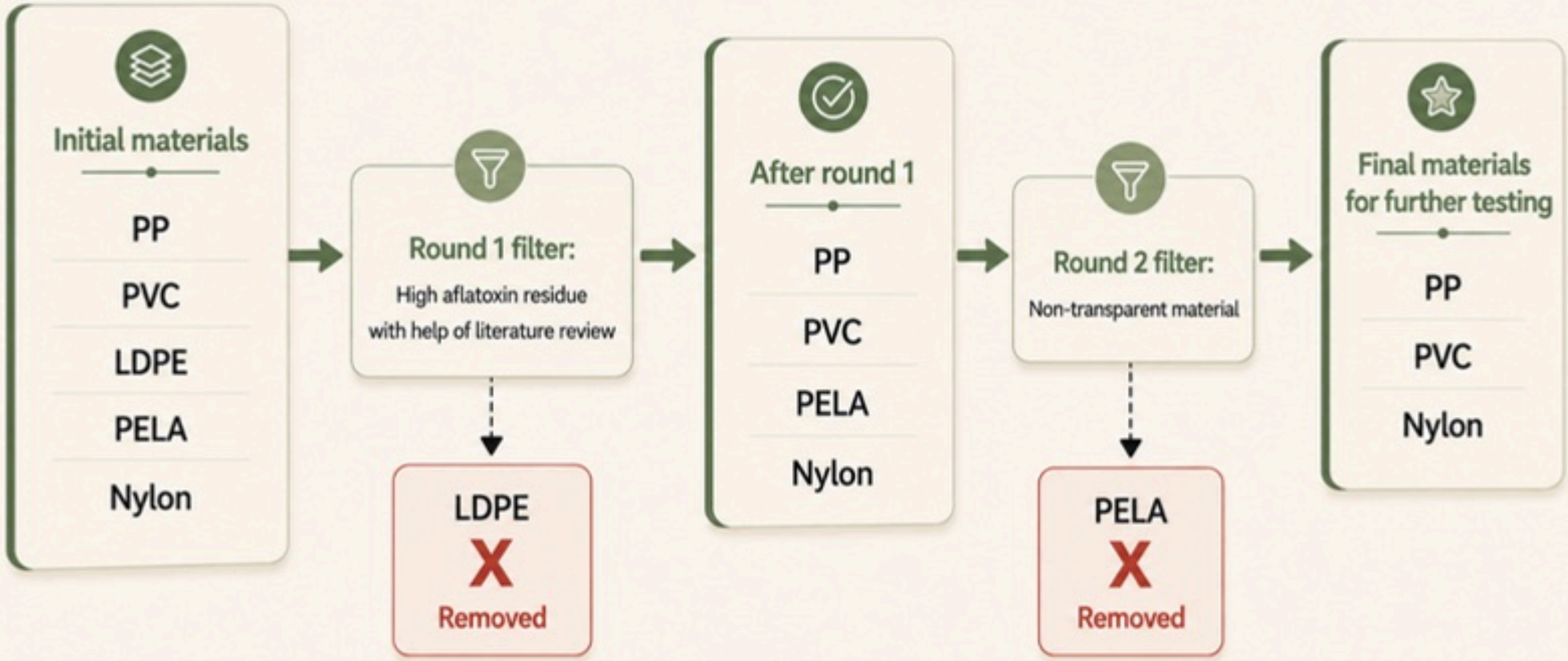
   Problem: Need to find **up-to-date field data** on aflatoxin

   - Rich Stutt (plant epidemiology), Gideon Darko (crop scientist) and Lara (Computational Epidemiologist)
   - Rich directed us to the correct references for literature review: similar experiment for Aspergillus Flavus in peanuts
   - Laura advised us on modelling
   - Gideon advised us on possible experiment set up (Johndeere device for moisture, Reveal Q for aflatoxin), and provided data for maize aflatoxin level post harvest

| KEJETIA | | |
|---|---|---|
| Sample ID | Variety | Aflatoxins Levels |
| KJG1 | Gold | 14.3 |
| KJG2 | Gold | 18 |
| KJG3 | Gold | 16.9 |
| KJG4 | Gold | 15.8 |
| KJG5 | Gold | 14.8 |
| KJG6 | Gold | 15.3 |
| KJG7 | Gold | 16.3 |

**Maize aflatoxin level post-harvest, with <1 being EU safety standard**

# Current Progress: Scope of materials



# Sodium Hypochlorite-surface Interaction

# Sodium Hypochlorite Experiment

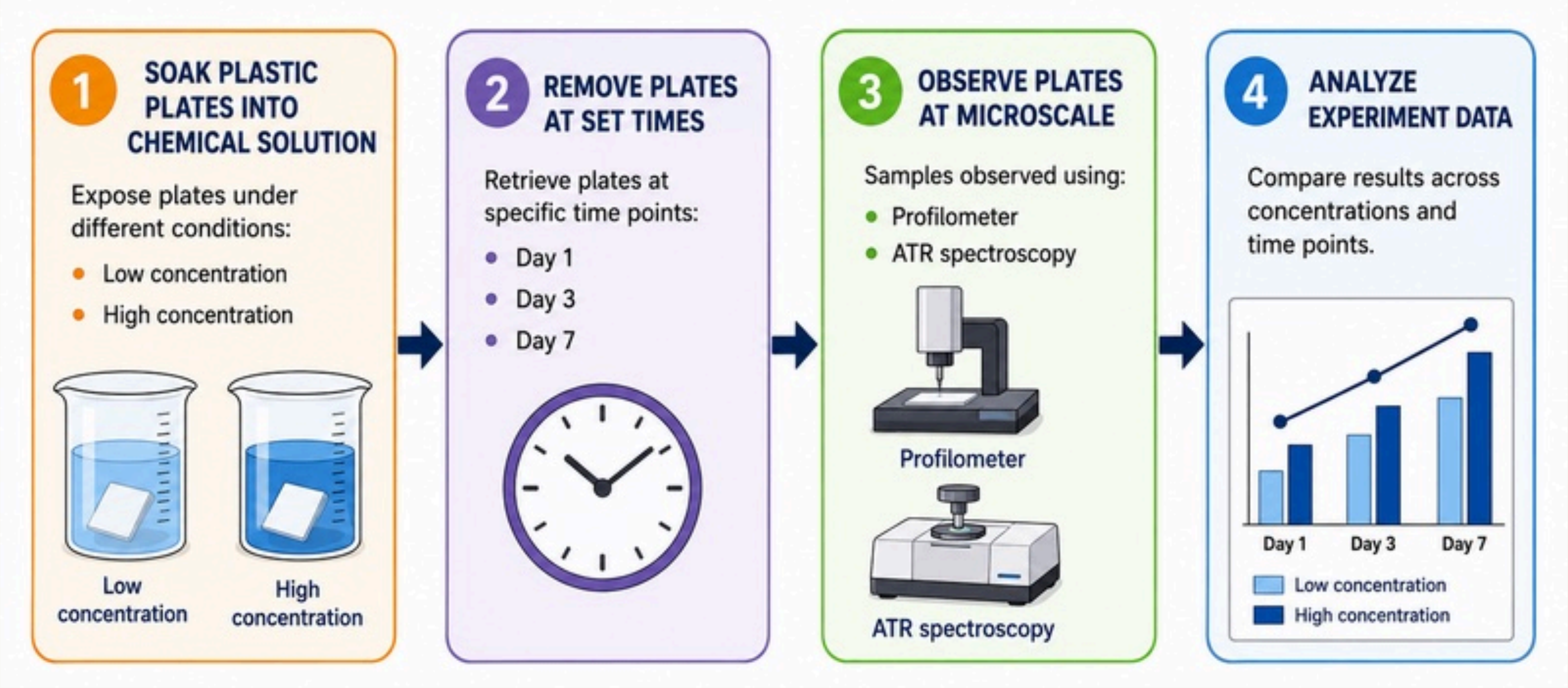
## Theory

- Different plastics have **different resistance to NaOCl**
- NaOCl is to cover the outside of the container, plastics which retain integrity better suited, less reactive

## Experiment Information

- Lab supervisor: Eugene Terentjev from the plastic lab
- Expected observation: macro and micro changes in plastic. (ATR and prolifometer and microscopy)

# Experiment Procedure



# EXPERIMENT SET UP

## 1 PP

potential candidate

## 2 PVC

currently used in drying containers

## 3 Nylon

potential candidate

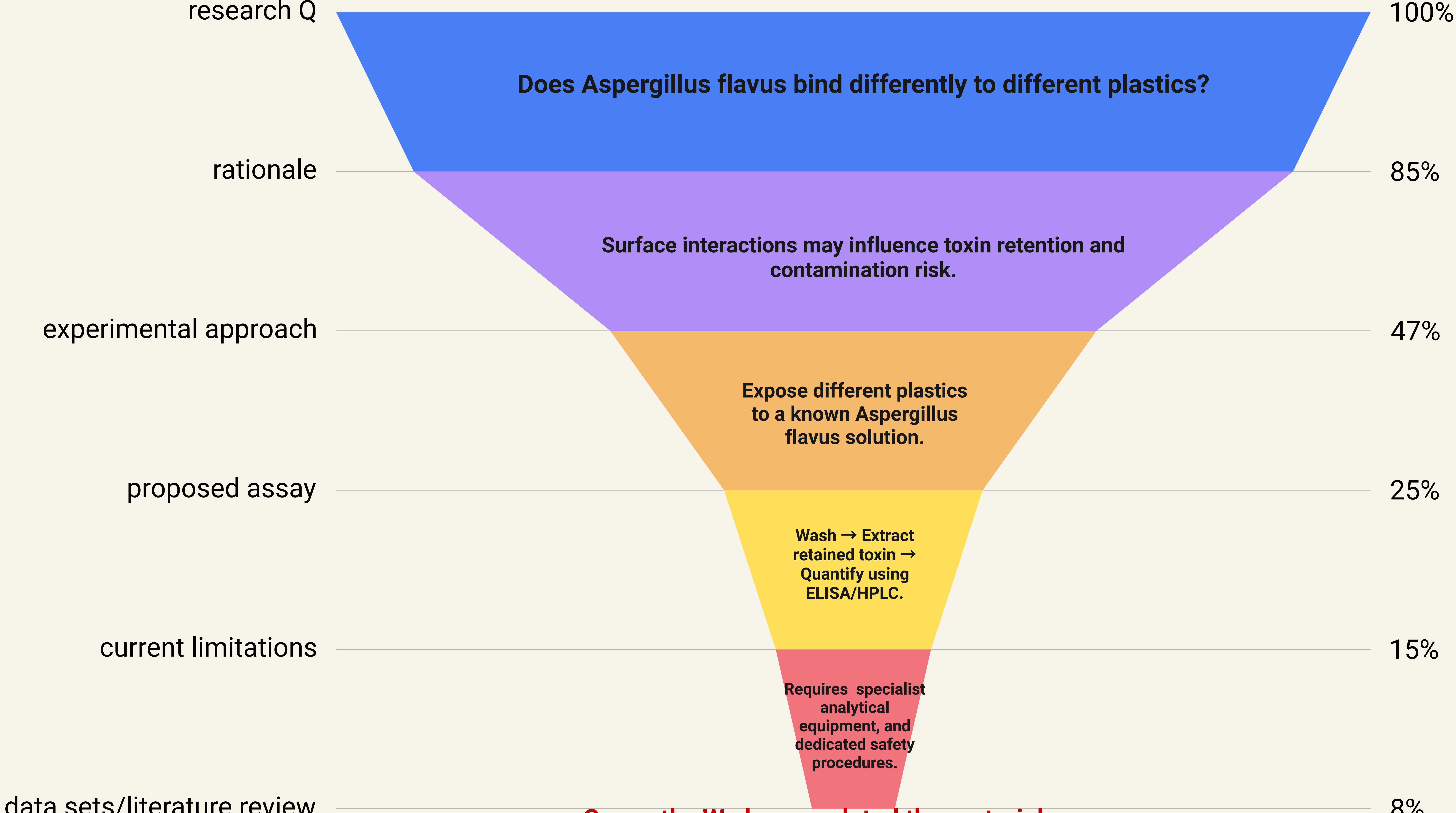
- prep each plastic (cut into small equal pieces)

- soak in NaOCl
- in high and low conc (dilute with water to create buffer)

- 1 without treatment
- 1 at higher temp (water bath) for ageing (high conc)
- 1 high conc
- 1 low conc

measuremnts
- ATR spectrography
- microscropy
- prolifometer
- assess surface changes, visual changes thickness, pitting, etc

# Aspergillus flavus-surface Interaction




**Currently: We have updated the material scope.**

**In next week: Additional research data is still needed to finalise the conclusion.**

# Team Improvements

## Kunran

- Improved professional communication with academics and lab staff
- Improved experimental design thinking
- Improved literature review skills

## Mirha

- Improved professional communication with academics and lab staff
- Improved understanding of materials testing in microscale
- Improved ability to learn unfamiliar technical content quickly

# Future Plans

### June 2th

**Visit materials lab** for experiment preparation

### June 3th

Received ordered materials, continue aflatoxin related **literature review**

### June 4th

**Conduct exposure test** for Sodium Hypochlorite at Eugene's plastic lab

### June 5th

Oxygen & moisture experiment

### Monday

**Collect results** from plastic lab

# Future Challenges to Explore

# Future Challenges to Explore

1. Oxygen & Moisture Distribution Research/Experiments

   Does the **non-uniform distribution** of O2/moisture relate to bag material?


Oxygen and moisture inside the bag can affect growth conditions.

2. Aflasafe (natural biocontrol product) for Kenya

   Can we develop a biocontrol product for Kenya to **prevent pre-harvest corn infection**?

# Experiment Procedure

## Detect non-uniform Oxygen & Moisture Distribution for different materials

Measurements

Bags

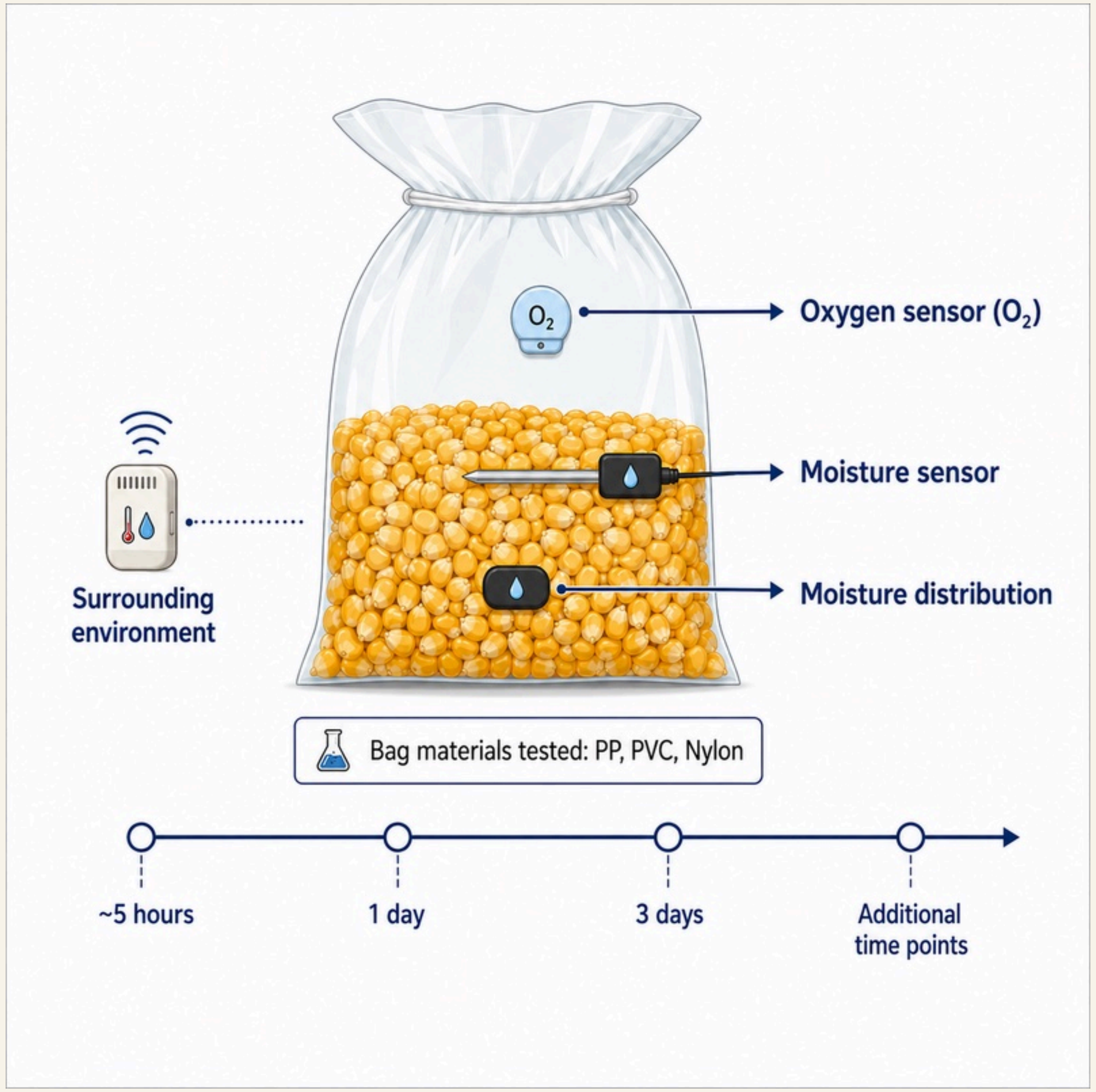
- Make bags using plastic films (PP, PVC, nylon) use **heat sealer,** replicating drying container

Moisture

- Spatial moisture distribution within the maize using John Deere sensors and sensors around inside bag.

Oxygen

- Oxygen concentration measured at multiple locations throughout storage.
- Oxygen depletion can be used as an indicator of $CO_2$ production.



# Thank You!